Hybrid ALNS–Quantum Optimization of Phase-Coded Radar Waveforms

An Adaptive Large Neighbourhood Search Framework with Quantum Repair

Algorithm Design Dossier

VerneX Quantum Algorithm Challenge 2026

Proposed algorithm. Adaptive Large Neighbourhood Search (ALNS) controls full-length binary-phase waveform optimization ($Q = 2$), while a quantum optimizer is tasked with selected, smaller LABS neighbourhoods. Classical candidate generation, acceptance and descent provide a robust fallback. We give an exact analytical QUBO reduction and a direct higher-order Ising formulation for quantum execution, with a pathway to polyphase codes ($Q > 2$). Implemented classical and landscape experiments motivate the design; quantum repair performance remains to be validated.

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Supporting code and experiments: `https://github.com/Eulerq-Software/Vernex-Challenge-Submission`

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1 Executive Summary

Phase-coded radar waveform design seeks sequences with small autocorrelation sidelobes, supporting high-quality synthetic aperture radar (SAR) imaging. The supplied Thales use case specifies a matched-filter, two-phase validation problem ($Q = 2$), equivalent to the Low Autocorrelation Binary Sequences (LABS) problem [1]. Its objective has long-range, higher-order interactions and an exponentially large configuration space.

Our proposed solution is a hybrid ALNS–quantum algorithm. Adaptive Large Neighbourhood Search (ALNS) organizes the optimization of the complete waveform: it builds and refines incumbents, selects informative subsets of phase variables, evaluates exact full-length objective values and controls acceptance. A quantum optimizer or sampler provides an additional *repair operator* on selected small subproblems. Quantum-generated candidates are reconstructed, locally relaxed and compared against the best classical result. This focuses costly quantum computation where it can be most useful while retaining a functional classical route at every stage.

A fully quantum treatment of large LABS instances is not the most practical near-term execution plan. Direct higher-order quantum circuits may involve many dense, nonlocal interactions, while the equivalent unconstrained QUBO requires auxiliary variables, additional couplings and penalty scales. Current hardware constraints and the cost of full-length quantum execution therefore motivate restricting the quantum portion; this is a feasibility argument, *not a proof that quantum-only approaches are impossible*. We propose that hybrid optimization is a promising route to practical performance, with its actual benefit to be determined experimentally.

The classical foundation already exists. We implemented several solver families, including Tabu search, variable neighbourhood search (VNS) and ALNS, and studied why improvement often stagnates. Experiments reveal that reaching a better local minimum can require an initially uphill move followed by a productive descent cascade. For several tested sizes and starting states, sequentially guided perturbations reached improved basins approximately 2.8–6.7 times as often as random perturbations of the same nominal size. These observations guide neighbourhood selection and suggest that a quantum repair operator should be judged by *post-relaxation improvement*, not solely immediate energy reduction.

For $Q = 2$, classical solvers evaluate the original binary-spin objective directly. We additionally derive an exact auxiliary-variable QUBO for quantum annealing or other quadratic-Ising methods. Gate-model QAOA can instead use the restricted higher-order Ising Hamiltonian without a full QUBO expansion. We outline an extension to $Q > 2$ by changing the phase encoding and generalizing the neighbourhood operators.

Proposal and evidence. The principal algorithm proposed in this submission is ALNS with quantum-assisted neighbourhood repair. The classical solvers and landscape measurements have been implemented and experimentally studied; the quantum repair component and the multi-phase extension are specified for implementation and matched-budget validation. No demonstrated quantum advantage or completed LABS-QPU result is claimed.

1.1 Contributions and maturity

Element	Contribution	Current status
Classical solvers	Seven general classical baselines tested through $N = 1000$; ALNS destroy/repair variants explored.	**Implemented / tested**
Landscape characterization	Direct neighbours, basin escapes, small-k triggers, cascades, and hard incumbents.	**Experimentally studied**
Guided perturbations	Sequential dynamic-bit selection improved escape success in several matched experiments.	**Experimentally supported**
Binary QUBO	Exact auxiliary-variable quadratization of the matched $Q = 2$ objective.	**Analytical formulation**
ALNS–quantum repair	Quantum optimization of selected ALNS neighbourhoods; reconstruct and relax measured candidates classically.	**Proposed algorithm / to validate**
Polyphase extension	$Q > 2$ phase encoding, generalized objective and binary reductions.	**Proposed / to benchmark**

Table 1: Scope and status of the main components.

2 Industrial Context, Objective and Scope

The Thales use-case concerns optimizing phase-coded waveforms for airborne SAR. A pulse is divided into N chips, each assigned a phase from a finite alphabet of size Q. Correlation of the received echoes with a reference waveform produces a main peak and sidelobes. The quality criterion in the provided problem specification is the sum of squared nonzero-delay correlations, referred to there as the ISLR optimization objective [1, 2].

Let $q_i \in \{0, \dots, Q-1\}$ and

$$x_i = \exp\left(\frac{2\pi \mathrm{i}\, q_i}{Q}\right), \qquad i = 1, \dots, N. \tag{1}$$

For matched filtering, define the aperiodic autocorrelation

$$\rho_k(\mathbf{x}) = \sum_{i=1}^{N-k} x_i \overline{x_{i+k}}, \qquad k = 1, \dots, N-1, \tag{2}$$

and the objective

$$\min_{q_1,\dots,q_N \in \{0,\dots,Q-1\}} E_Q(\mathbf{x}) = \sum_{k=1}^{N-1} |\rho_k(\mathbf{x})|^2. \tag{3}$$

The zero-lag correlation, equal to N for unit-modulus codes, is omitted. The use-case specifies validation for $Q = 2$ at $N \in \{5, 7, 11\}$ using Barker codes and gives three headline indicators: optimization time versus N, largest code length tested, and objective relative to the Barker reference $(N-1)/2$ [1].

2.1 Binary-phase specialization

For $Q = 2$, $x_i = s_i \in \{-1, +1\}$ and

$$C_k(\mathbf{s}) = \sum_{i=1}^{N-k} s_i s_{i+k}, \qquad E(\mathbf{s}) = \sum_{k=1}^{N-1} C_k(\mathbf{s})^2. \tag{4}$$

This is the LABS objective, also known as the energy of the aperiodic Bernasconi model [3, 4]. Its search space contains 2^N binary sequences before symmetry reductions. For a reported energy $E > 0$, the merit factor is

$$F(\mathbf{s}) = \frac{N^2}{2E(\mathbf{s})}, \tag{5}$$

so minimizing E is equivalent to maximizing F at fixed N. We will also report

$$R_{\text{Barker}}(N) = \frac{E(\mathbf{s})}{(N-1)/2} = \frac{2E(\mathbf{s})}{N-1}, \tag{6}$$

following the challenge's requested reference convention. This reference *is not an optimality ratio* for arbitrary N: its denominator is a Barker benchmark, not the best-known energy at length N.

2.2 Assumptions and boundaries

Assumption / boundary	Implication for this submission
Matched filtering $y = x$	The reported experiments minimize only nonzero-delay autocorrelation energy, not mismatched-filter SNR trade-offs.
Two phases in tested algorithms	All numerical landscape and classical solver results concern $Q = 2$.
Equal-duration, unit-modulus chips	No amplitude or continuous-phase optimization is claimed.
Discrete unconstrained code design	No deployment-specific spectrum, peak-power or transmitter-hardware restrictions are modeled.
Classical benchmark evidence	Quantum performance, QPU runtime and noise resilience remain to be evaluated.

Table 2: Assumptions register and validity limits.

3 Exact QUBO Reformulation of the $Q = 2$ Problem

Our implemented classical LABS algorithms evaluate the original binary-spin objective directly; they do not use the unconstrained QUBO constructed below. The original objective is *not itself quadratic*: expanding eq. (4) produces two-body and four-body interactions. For quantum optimizers requiring a quadratic objective, we analytically derive an *exact* QUBO using product auxiliaries and consistency penalties [5]. A separate Gurobi implementation in the code repository uses auxiliary XOR variables with explicit constraints in a constrained quadratic model; that is different from an unconstrained QUBO.

Write $s_i = 1 - 2b_i$, where $b_i \in \{0, 1\}$. For every pair $i < j$ introduce $t_{ij} \in \{0, 1\}$ intended to equal $b_i b_j$. With these variables,

$$\widetilde{C}_k(\mathbf{b}, \mathbf{t}) = (N-k) - 2\sum_{i=1}^{N-k}(b_i + b_{i+k}) + 4\sum_{i=1}^{N-k} t_{i,i+k}. \tag{7}$$

If all product identities hold then $\widetilde{C}_k = C_k$; moreover, $\sum_k \widetilde{C}_k^2$ is *quadratic* in the binary variables because each $\widetilde{C}_k$ is affine. The product condition can itself be encoded by the quadratic penalty

$$P_{ij} = b_i b_j - 2b_i t_{ij} - 2b_j t_{ij} + 3t_{ij}. \tag{8}$$

For binary arguments $P_{ij} \geq 0$ and $P_{ij} = 0$ exactly when $t_{ij} = b_i b_j$. An exact unconstrained quadratic objective is therefore

$$\boxed{\min_{\mathbf{b},\mathbf{t}\in\{0,1\}} H_\lambda(\mathbf{b}, \mathbf{t}) = \sum_{k=1}^{N-1} \widetilde{C}_k^2 + \lambda \sum_{1\leq i<j\leq N} P_{ij}.} \tag{9}$$

This expression is of QUBO form $\mathbf{z}^\mathsf{T} A \mathbf{z} + c$ for $\mathbf{z} = (\mathbf{b}, \mathbf{t})$. The additive constant does not change the optimum. An explicit sufficient (but conservative) exactness condition is

$$\lambda > U_N := \sum_{k=1}^{N-1} (N-k)^2 = \frac{N(N-1)(2N-1)}{6}. \tag{10}$$

Indeed, every feasible sequence has $E \le U_N$, whereas a violated product identity contributes at least λ and all squared terms are nonnegative. Thus no infeasible assignment can minimize eq. (9).

The full construction uses $N + \binom{N}{2}$ binary variables; naive expansion may create $O(N^3)$ quadratic interaction contributions before coefficient aggregation. The sufficient penalty in eq. (10) may also create an impractically wide coefficient scale. These are concrete reasons *not* to assume that a full-length QUBO maps efficiently to current quantum hardware. In practice we would investigate tighter instance-specific penalties, sparse representations, direct higher-order Hamiltonians and restricted neighbourhoods.

3.1 Direct higher-order Hamiltonian alternative

Using $s_i \leftrightarrow Z_i$, the original energy yields a diagonal Pauli Hamiltonian

$$H_C = c_0 I + \sum_{i<j} J_{ij} Z_i Z_j + \sum_{i<j<k<\ell} K_{ijk\ell} Z_i Z_j Z_k Z_\ell. \tag{11}$$

Some nominal four-body products reduce to constants or two-body terms when indices coincide. Gate-model QAOA can phase-separate directly with eq. (11), avoiding the QUBO auxiliary qubits, but circuit synthesis and connectivity may be expensive. Conversely, hardware designed for quadratic Ising/QUBO optimization motivates eq. (9); the best representation is execution-pathway dependent.

4 Classical Solver Framework

Our first phase of work established multiple fully classical algorithms for the binary-phase problem. The evaluated families were greedy search, random search, coordinate descent, genetic algorithms, simulated annealing, Tabu search and VNS. These provide useful local, stochastic and metaheuristic reference points; specialized LABS algorithms from the literature remain an additional comparison target [4].

We implemented an ALNS framework with random, contiguous-block, best-flip, correlation-guided and mixed destroy operators. Classical repair choices include greedy and randomized greedy reconstruction, exact small-neighbourhood enumeration, and simulated annealing. These classical components form the *controller and fallback* of our proposed hybrid solver. Quantum neighbourhood repair is added as a complementary operator, not as a replacement for full-length ALNS search. Its adoption is decided by measured improvement per unit end-to-end computational cost.

A repeated observation was that ordinary improving repair from strong local minima easily stagnated. For example, in one $N = 200$ simulated annealing trajectory the energy changed approximately $4456 \to 4648 \to 4476$: the search escaped uphill but failed to beat the incumbent. This is not a general failure theorem for simulated annealing; it motivated deeper empirical study of the basin structure and of which perturbations lead to genuinely better solutions.

4.1 Sequentially guided perturbation operator

For an incumbent one-flip local minimum $\mathbf{s}^*$, a uniform exact-d perturbation chooses d distinct positions, flips them and applies greedy descent. Our sequentially guided alternative makes the selection *adaptive to the intermediate configurations*: compute all single-flip energy changes, select stochastically among the five lowest-delta admissible positions, flip the selected bit, recompute the deltas, and continue until d distinct positions have been changed. Previously selected positions are not chosen again during the same perturbation. Finally, deterministic best-improvement descent is applied and the full objective is compared with the starting incumbent.

The operator deliberately allows uphill intermediate moves. Its guidance is intended to build a trajectory whose changing correlations make subsequent rearrangements more productive. These are empirical design choices rather than a guarantee of improvement, and their effectiveness must

be checked against uniformly random perturbations and other strong local-search strategies under comparable budgets.

5 Empirical Landscape Characterization

We studied the energy landscape around strong one-flip local minima to understand whether the main difficulty was discovering a better nearby binary string or reaching a better *basin of attraction*. For a selected set of flipped positions S, let $\mathbf{s} \oplus S$ denote bit flips at precisely those indices and define Relax as best-improvement single-bit descent to a local minimum. A successful basin transition satisfies

$$E(\mathrm{Relax}(\mathbf{s}^* \oplus S)) < E(\mathbf{s}^*), \tag{12}$$

even when $E(\mathbf{s}^* \oplus S) > E(\mathbf{s}^*)$.

5.1 Escape is a two-stage process

Random exact-d perturbations around selected local minima of lengths $N = 50, 100, 200$ were almost uniformly worse *before* descent in the reported experiments (10,000 samples at several distances). Nevertheless, some relaxed perturbations reached lower-energy local minima. Around roughly 10% Hamming distance, leaving the original basin was common, while reaching a *better* basin remained uncommon. At $N = 200$, selected random 15% perturbations found a better basin only about 0.04–0.18% of the time for some strong incumbent states.

This evidence suggests that simple escape probability and productive escape probability must be distinguished. We therefore focus on the conditional outcome after local relaxation, not only the energy of the immediately perturbed state. Quantum candidates will be assessed by the same end-to-end measure, preventing a quantum component from appearing successful merely because it explores many high-energy configurations.

5.2 Small triggers and downhill cascades

We exhaustively enumerated small k-bit perturbations around selected local minima and then relaxed each candidate. At $N = 50$, none of the tested one- or two-bit perturbations gave a productive basin transition, while three-bit productive triggers existed around all three investigated local minima. The tested direct perturbations did not themselves lower the energy, yet some of them initiated longer downhill cascades. At $N = 100$ productive $k = 3$ triggers were rare in some cases; at $N = 200$ a rare $k = 2$ productive trigger occurred for one tested incumbent.

This motivates the diagnostic quantity

$$k^*(\mathbf{s}) = \min\{|S| : \ E(\mathrm{Relax}(\mathbf{s} \oplus S)) < E(\mathbf{s})\}, \tag{13}$$

when the set is nonempty. A small trigger may cause a much larger final Hamming displacement after descent. Consequently, the most useful neighbourhood operator need not locate an immediate lower-energy string; it may locate a configuration that produces a large and beneficial cascade.

5.3 Dynamical diagnostics

Successful and unsuccessful cascades differed more clearly in their subsequent descent than in static properties of the initial perturbation. For example, around $N = 200$ successful cascades averaged approximately 32–33 descent steps versus about 24 for failures in the reported sample. At a checkpoint near $0.15N$ descent steps, only 11–12% of all tested candidates remained active, but 77–96% of eventually successful trajectories were retained among them. This implied a roughly 7–9-fold enrichment in successful candidates.

The checkpoint analysis is a diagnostic, not a deployed machine-learning component. It suggests a possible low-cost screening or early-allocation strategy: invest additional refinement resources

in candidates that exhibit persistent productive descent. A quantum sampler would need to improve this end-to-end selection process, rather than just replicate ordinary short-range energy minimization.

6 Proposed Algorithm: ALNS with Quantum Neighbourhood Repair

Design principle. Use ALNS to search the complete LABS space and delegate selected difficult local subproblems to a quantum optimizer. The quantum routine is a *repair operator inside ALNS*, rather than a stand-alone solver for the entire waveform. The ALNS controller selects destroyed variables and generates candidates, assigns classical or quantum repair under an explicit resource budget, and applies common relaxation and acceptance logic. It updates operator-selection statistics using the quality of the resulting *complete* waveform.

At an incumbent $\mathbf{s}$, ALNS selects a subset $S \subseteq \{1, \ldots, N\}$ of m active phase variables, with $m \ll N$; the remaining spins are frozen. The exact restricted energy is

$$E_S(\mathbf{u} \mid \mathbf{s}) = E(\mathbf{s}_{\bar{S}}, \mathbf{u}_S), \qquad \mathbf{u}_S \in \{-1, +1\}^m. \tag{14}$$

Substituting the fixed spins gives a cost polynomial of degree at most four on m logical variables. Quantum repair samples candidate assignments $\mathbf{u}_S$; ALNS restores them to full-length waveforms, applies classical local descent, evaluates the original LABS energy and accepts improvements under the chosen ALNS acceptance rule. Classical repair is always available when quantum execution is too costly or ineffective.

6.1 Why not solve the entire LABS instance quantum-mechanically?

A full-length direct Ising formulation requires N logical spins and a potentially dense collection of higher-order cost terms; a full QUBO can require $N + \binom{N}{2}$ binary variables before hardware embedding. For sufficiently large N, coherence, connectivity, gate depth, qubit count, annealer embedding and shot/optimization overhead are substantial practical obstacles. Even a successful quantum subroutine must compete with mature classical heuristics that evaluate LABS rapidly. Consequently, we *do not assume* that a quantum-only implementation is currently efficient at application-relevant sizes. This does not imply that full quantum approaches are impossible; it motivates the concrete hybrid architecture in which m, rather than N, determines the size of each quantum optimization call.

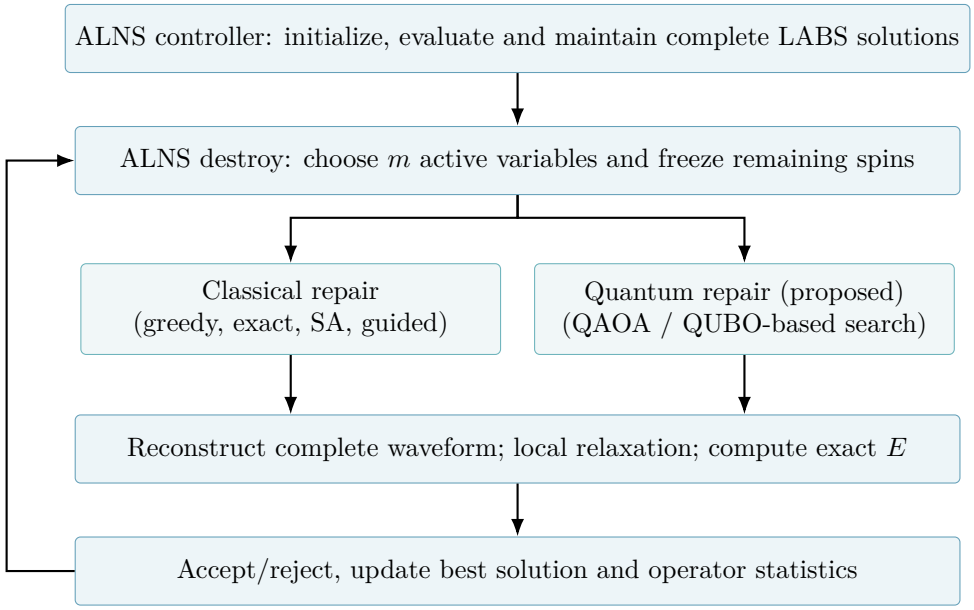


Figure 1: The proposed ALNS–quantum algorithm: the implemented classical ALNS framework is the global controller; quantum repair is the specified, yet-to-be-benchmarked neighbourhood operator.

6.2 Gate-model execution: QAOA on a restricted Hamiltonian

Let $H_{C,S}$ be the diagonal Pauli-Z operator corresponding to eq. (14), after combining repeated spin monomials. A baseline QAOA circuit uses the m-qubit state

$$|\psi(\boldsymbol{\gamma},\boldsymbol{\beta})\rangle = \prod_{r=1}^{p}\left[e^{-\mathrm{i}\beta_r H_M} e^{-\mathrm{i}\gamma_r H_{C,S}}\right]|+\rangle^{\otimes m}, \qquad H_M = \sum_{j\in S} X_j, \tag{15}$$

where the rightmost operator acts first. The angles are optimized classically with a bounded number of evaluations; measurement samples supply possible new neighbourhood assignments. A k-local diagonal term $c\,Z_{i_1}\cdots Z_{i_k}$ is implemented by a parity-computation network and a corresponding Z rotation, subject to connectivity and gate-compilation costs. We do not assume this dense cost Hamiltonian has constant-depth implementation.

A key issue is that optimizing the *immediate* restricted energy E_S does not necessarily find productive uphill triggers. We therefore specify two trainable quantum-repair variants. The *energy-trained* variant optimizes the conventional QAOA expectation $\langle H_{C,S}\rangle$; it is a necessary baseline. The *basin-trained* variant uses the same implementable restricted Hamiltonian but evaluates measured candidates after classical descent. For circuit parameters $\theta = (\boldsymbol{\gamma},\boldsymbol{\beta})$ and measurement distribution P_θ, its classical outer-loop objective can be

$$L_{\text{basin}}(\theta) = \mathbb{E}_{\mathbf{u}\sim P_\theta}[E(\text{Relax}(\mathbf{s}_{\bar{S}}, \mathbf{u}_S))]\,. \tag{16}$$

This scoring requires *classical* full-waveform evaluations and descent after measurement; it does not assume a coherent oracle for the descent map. We will also compare alternate mixer choices, initialization states and sample-diversity controls. Success must be judged by final post-relaxation quality, taking all quantum and classical costs into account.

6.3 Quadratic-Ising / annealing pathway

The exact QUBO of eq. (9) provides an alternative to direct higher-order gate-model execution. A selected neighbourhood can be quadratized and submitted to a quadratic-Ising optimizer or a quantum annealer, provided the auxiliary count, coefficient dynamic range, connectivity and embedding overhead fit the chosen resource budget. Restricting the active original variables

can reduce the required auxiliaries substantially relative to a full N-chip formulation; however, affected correlations with frozen bits must remain in the cost. The selected execution route will be determined by measured resources, not by assuming equivalence of hardware or runtimes.

Algorithm 1 Proposed ALNS–quantum hybrid solver (design specification)

Require: Length N, time budget B, initial code $\mathbf{s}_0$, admissible quantum widths $\mathcal{M}$, repair portfolio $\mathcal{O}$

$\mathbf{s} \leftarrow \mathrm{Relax}(\mathbf{s}_0)$; $\mathbf{s}_{\mathrm{best}} \leftarrow \mathbf{s}$
while budget remains **do**
  ALNS destroy: select operator and active set S, with $|S| = m$
  Construct exact restricted objective $E_S(\cdot \mid \mathbf{s})$
  Repair choice: select a classical repair operator or, if $m \in \mathcal{M}$ and resource gating permits, quantum repair
  Generate candidate restricted assignments $\mathcal{U}$
  for each $\mathbf{u} \in \mathcal{U}$ **do**
    Restore $\mathbf{y} \leftarrow (\mathbf{s}_{\bar{S}}, \mathbf{u}_S)$
    Apply classical descent $\mathbf{z} \leftarrow \mathrm{Relax}(\mathbf{y})$
    Evaluate exact full-waveform energy $E(\mathbf{z})$
    if $E(\mathbf{z}) < E(\mathbf{s}_{\mathrm{best}})$ **then**
      $\mathbf{s}_{\mathrm{best}} \leftarrow \mathbf{z}$
    end if
    Update incumbent $\mathbf{s}$ by the selected ALNS acceptance rule
  end for
  Update repair scores using post-descent gain and total time
end while
return $\mathbf{s}_{\mathrm{best}}$ and full experimental log

6.4 When would quantum assistance be useful?

The proposed ALNS–quantum method is useful only if quantum repair improves final energy under a fixed end-to-end budget, raises the rate of productive basin transitions per unit time, or improves scaling relative to strong classical repair methods. The quantity of interest for incumbent $\mathbf{s}$ and candidate generator G is

$$p_{\mathrm{improve}}(G \mid \mathbf{s}) = \Pr_{\mathbf{y}\sim G}[E(\mathrm{Relax}(\mathbf{y})) < E(\mathbf{s})]\,. \tag{17}$$

This must be accompanied by time for Hamiltonian construction, compilation, shots, parameter optimization, communication, classical postprocessing and descent. The landscape observations provide a *motivation* for testing quantum search, but they do not by themselves establish tunnelling, an optimization speedup or a quantum-over-classical crossover.

7 Extension Beyond Two Phases ($Q > 2$)

The restriction $Q = 2$ was a deliberate starting point: it provides a well-studied binary reference problem, exact Barker validation cases, a direct spin representation and reproducible comparisons with classical LABS methods. It is not a limitation of the overall search architecture. For general Q, the waveform remains $x_i = \omega^{q_i}$, where $\omega = e^{2\pi \mathrm{i}/Q}$, and the objective remains eq. (3); the neighbourhood operators change a chip's phase rather than simply flipping its sign.

7.1 Exact one-hot encoding

Introduce $u_{i,a} \in \{0,1\}$ for chip i and phase index $a \in \{0,\dots,Q-1\}$, with

$$\sum_{a=0}^{Q-1} u_{i,a} = 1, \qquad x_i = \sum_{a=0}^{Q-1} \omega^a u_{i,a}. \tag{18}$$

Then the autocorrelation is

$$\rho_k = \sum_{i=1}^{N-k} \sum_{a,b=0}^{Q-1} \omega^{a-b}\, u_{i,a} u_{i+k,b}. \tag{19}$$

Each $|\rho_k|^2 = (\Re\rho_k)^2 + (\Im\rho_k)^2$ is a real binary polynomial of degree at most four. Introducing product variables $v_{i,k,a,b} = u_{i,a}u_{i+k,b}$ makes the real and imaginary parts of ρ_k linear in v. Quadratic AND penalties of the same form as eq. (8), together with quadratic one-hot penalties $\eta \sum_i (1 - \sum_a u_{i,a})^2$, produce an exact QUBO when the penalties are sufficiently large. The formulation can require $NQ + O(N^2Q^2)$ binary variables and should therefore be applied to resource-limited subproblems rather than indiscriminately to full codes.

7.2 Compact phase encoding and search operators

A second route uses $r = \lceil \log_2 Q \rceil$ binary variables per phase index, with constraints forbidding unused bit strings when Q is not a power of two. This reduces primary variable count but can introduce higher-order interactions and more complex phase-dependent circuit synthesis. For the first extension, $Q = 4$ is attractive because each phase index has a natural two-bit encoding. We propose evaluating one-hot QUBO, compact binary encoding, and direct phase-index search under the same objective and computational budget.

At the algorithm level, binary flips become phase-reassignment moves, block and correlation-guided destroy operators remain meaningful, and classical relaxation searches among $Q-1$ alternative phases at each position. The $Q = 2$ findings about productive cascades are hypotheses to test in the polyphase setting, not results that can be transferred without evidence [7]. The mismatched-filter case adds a separate complex replica with SNR constraints and is considered an optional subsequent extension, not part of the presently validated scope.

8 Completed Experiments and Results

The following tables summarize experiments carried out during our classical research phase. They are included to document completed work and to justify the algorithmic design. Where the historical summary does not record processor model, exact wall-clock budget or run-count for a particular aggregate table, those details are not inferred. Consequently, these results are *internal baselines and diagnostic measurements*, not a controlled claim of superiority over published state-of-the-art specialized LABS solvers.

8.1 Barker-code correctness checks

An outside-in beam-search construction, one of the explored classical approaches, reproduced the matched-filter optimum energies for the three requested Barker validation sizes. This validates the objective convention on these test cases, although it does not validate the quantum subroutine (which is proposed separately).

N	Q	Known optimum E	Obtained E (beam search)
5	2	2	2
7	2	3	3
11	2	5	5

Table 3: Correctness results for the Barker-code tests defined by the Thales problem statement.

8.2 General classical solver benchmarks

Table 4 shows representative *median energies* from our earlier solver benchmark sweep. Tabu search was effective at several short and medium sizes; VNS was markedly better than this particular Tabu implementation at the largest reported lengths. Neither method should be called best-in-class based on this table alone.

N	Tabu median E	VNS median E	Tabu $F^\dagger$	VNS $F^\dagger$
50	241	245	5.187	5.102
100	1,066	1,036	4.690	4.826
200	4,224	4,448	4.735	4.496
300	10,102	10,062	4.455	4.472
500	38,850	28,254	3.218	4.424
1000	209,636	115,956	2.385	4.312

Table 4: Representative historical median energies. $F^\dagger = N^2/(2E_{\text{median}})$ is computed from the displayed median energy and is *not* separately the median over trial-level merit factors.

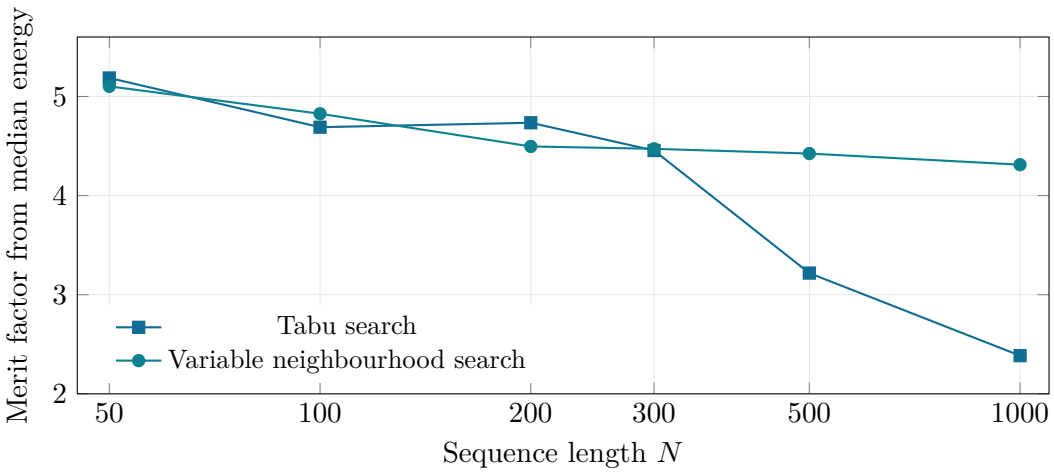


Figure 2: Merit factors calculated from the historical median-energy benchmarks, not a controlled equal-hardware/equal-time comparison.

8.3 Guided versus random basin escapes

Our strongest positive neighbourhood result came from selecting perturbation indices sequentially based on the current single-flip deltas. Table 5 reports the empirical fraction of attempts that reached a better local minimum after complete greedy descent. Across the selected ILS-starting states and perturbation sizes, the guided procedure increased success probability by approximately 2.8–6.7 times, although the magnitude depended on N and d.

N	Perturbation fraction	Random	Guided	Ratio
101	10%	0.6%	1.7%	2.83×
101	15%	0.3%	2.0%	6.67×
151	10%	0.6%	2.6%	4.33×
151	15%	0.2%	0.8%	4.00×
199	10%	1.6%	4.7%	2.94×
199	15%	1.7%	5.4%	3.18×
199	20%	1.1%	5.0%	4.55×

Table 5: Selected completed basin-escape experiments using ordinary iterated-local-search incumbents. Success means finding a strictly lower energy *after* local relaxation; rates are historical summaries.

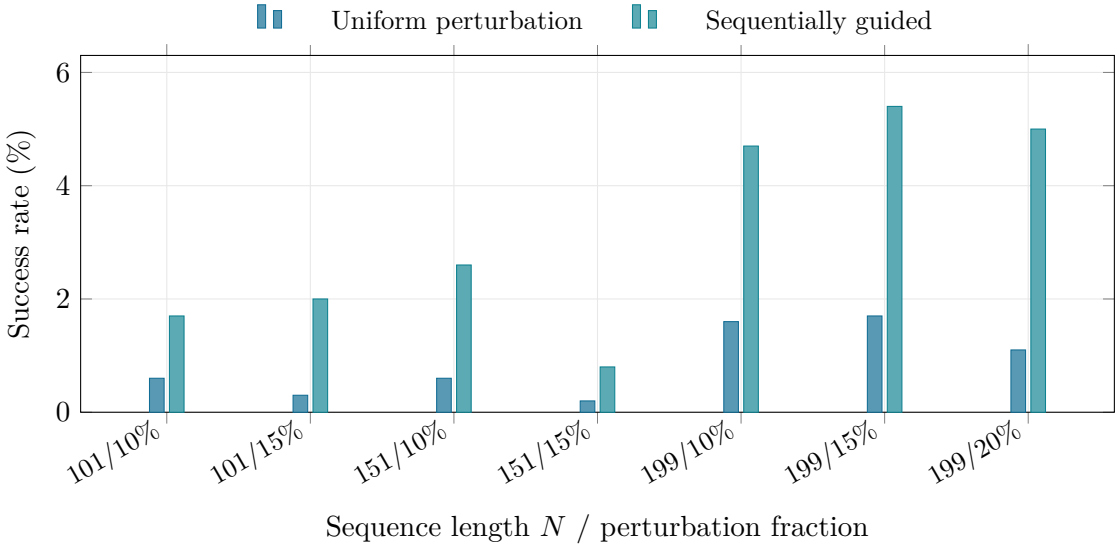


Figure 3: Higher basin-improvement success for the guided perturbation strategy on the selected benchmark configurations. Exact fractions are in table 5.

At $N = 199$, $d \approx 30$, successful uniform perturbations required about 30.3 additional greedy-descent flips and ended approximately 46.3 bits away from the initial state. Successful guided perturbations required only about 7.7 additional descent flips and ended about 30.8 bits away. These conditional statistics suggest that the guided construction itself performs much of the restructuring subsequently performed by a random-perturbation cascade.

8.4 High-quality Legendre starting points

We also examined best-rotated Legendre sequences at prime lengths. Representative initial energies and merit factors were:

N	E (best rotation)	Merit factor F
31	75	6.407
53	286	4.911
101	934	5.461
151	1,927	5.916
199	3,391	5.839

Table 6: Selected constructed starting points for challenging local-search experiments.

For each of $N = 101, 151, 199$, we tested 6,000 random and guided escape attempts across perturbation fractions of 10%, 15%, and 20%. No attempt improved its Legendre starting point: 0/18,000 across the three states. This does not establish that the states are globally optimal or that escaping them is impossible; it demonstrates the difficulty of these incumbents for the *tested* operators. At $N = 101$, a separate guided-search run from a poorer local minimum found $E = 922$, better than the Legendre starting value $E = 934$, emphasizing that good solutions may occupy basins that are difficult to reach from one another.

8.5 A qualification from later diagnostic tests

We do not claim universal superiority for every guidance heuristic. In a separate $N = 50$ diagnostic with 5,000 trials per strategy from a specified base state of energy 289, an activation-guided choice achieved 216 improvements (4.32%), whereas the tested best-single strategy achieved 219 improvements (4.38%). These variants and starting states are not interchangeable with the

sequential top-five method in table 5. Including a negative or near-neutral comparison helps identify which guidance mechanisms warrant further development.

9 Validation Plan and Criteria for Quantum Benefit

A quantum claim will be made only after testing the complete hybrid pipeline against competitive classical repair baselines. The primary comparison is *the quality of the final relaxed waveform under a fixed end-to-end computational budget*. Useful secondary outcomes are productive escape probability, expected number of successes per second, probability of improving the incumbent by a specified energy threshold, and scaling with N and quantum neighbourhood width m.

9.1 Controlled experimental design

For each length N, incumbent quality class and neighbourhood size m, all methods will receive the same starting waveforms and frozen variables. Candidate generators will include uniform perturbation, sequential guidance, classical small-neighbourhood enumeration (where tractable), classical stochastic optimization, energy-trained QAOA, basin-trained QAOA, and, where available, QUBO-based quantum or quantum-inspired optimizers. We will compare both stand-alone repair quality and the overall ALNS pipeline with and without quantum repair. We will separately compare equal candidate-evaluation counts and equal wall-clock budgets. Trial seeds, solver parameters and the full hardware configuration will be recorded.

We propose beginning with restricted problems small enough for exhaustive enumeration as a correctness and quality reference (for example $m \in \{8, 12, 16\}$, subject to actual resource feasibility), then testing larger restricted searches with fixed shot/iteration budgets. On a simulator, we can also compare sampled distributions before and after measurement-noise or gate-noise modeling. For any physical QPU results, wall-clock measurements will include queueing/communication when reporting user-facing latency, and clearly separate these effects from circuit execution time.

Metric	Interpretation and reporting rule
Final energy / merit factor	Best, median, spread and confidence intervals over independent runs at fixed N and budget.
Barker reference multiple	$2E/(N-1)$, labelled as a reference multiple, never as a generic optimality gap.
Post-descent improvement rate	Fraction satisfying $E(\mathrm{Relax}(\mathbf{y})) < E(\mathbf{s})$; report trial counts and confidence intervals.
Successes per second	Productive transitions divided by total end-to-end elapsed time, not quantum shots alone.
Classical–quantum ablation	Same selected S, same starting codes, and the same downstream relaxation/acceptance.
Resource footprint	Logical and physical qubits where applicable; Pauli-term count, gate depth, shot count, optimizer calls, QUBO density, ancillae and penalty scales.
Scaling	Quality versus time, largest validated N, and resource trends versus m and Q.

Table 7: Pre-specified evaluation metrics for assessing the proposed quantum component.

9.2 Decision rules for reporting results

We distinguish (i) a *demonstrated empirical advantage* on a specified benchmark, hardware and budget, (ii) *scaling evidence* without an observed crossover, and (iii) a *projected* advantage from resource or simulation models. No category is asserted by the current classical-only results. We will also report failures: if quantum candidate generation does not beat a strong classical baseline after compilation, shots and postprocessing, the correct conclusion is that this quantum method did not yield a benefit in the tested regime.

10 Feasibility, Constraints and Risk Mitigation

The primary technical trade-off is between expressive cost functions and executable quantum resources. Direct higher-order encoding can use m logical qubits for an m-bit restricted neighbourhood, but the cost Hamiltonian may contain many nonlocal interactions. A quadratic QUBO reduces interaction order at the cost of ancillary variables, large penalty coefficients and possibly dense connectivity. Increasing QAOA depth or sampling budget may also erase a candidate-quality gain once total computation time is counted.

We address these risks by keeping the quantum portion bounded and optional, by benchmarking exact restricted objectives against classical algorithms, and by retaining a reliable classical fallback. A resource gate will invoke the quantum branch only when the instance has an embeddable or compilable subproblem and when trial evidence suggests that the expected quality gain justifies its overhead. We also plan sensitivity tests for the neighbourhood-selection heuristic, objective penalty scales, parameter initialization, shot noise and local optimizer settings.

10.1 Constraints and risk register

Risk / limitation	Consequence	Proposed response
Quantum objective mismatch	Low immediate E need not imply productive post-descent escape.	Score after Relax; test alternate samplers.
Dense four-body Hamiltonian	High compilation depth/connectivity cost.	Restrict m; simplify terms; profile compilation.
QUBO auxiliary/penalty overhead	Embedding infeasible or poor numerical conditioning.	Use direct higher-order representation or smaller QUBOs.
Very strong incumbent basins	Low probability of any further improvement.	Stratify by incumbent merit factor; report null results.
Noise and optimizer overhead	Quantum sampled candidates may be inferior per unit wall time.	End-to-end budgets and classical fallback.
$Q > 2$ extrapolation	Binary landscape conclusions may not transfer.	Separate polyphase benchmark programme.

Table 8: Technical limitations and specific mitigation measures.

11 Development Roadmap and Integration Pathway

The proposed ALNS–quantum solver is specified here as an executable research plan with independently auditable checkpoints. We first freeze and validate the implemented ALNS/classical foundation, then implement restricted quantum repair, connect it to the ALNS controller and benchmark end-to-end performance. This sequence provides a practical hybrid route without presupposing that a fully quantum solver is viable at large N or that a quantum advantage has already been observed.

Phase	Activities	Measurable deliverables
I. Baseline freeze	Re-run correctness and representative multi-seed classical benchmarks; document hardware and wall time.	Barker validation, quality–time curves, reproducible configurations.
II. Quantum repair	Build exact restricted $H_{C,S}$ and QUBO; simulate energy- and basin-trained QAOA; validate with exact m-bit enumeration.	Hamiltonian checks, resource reports, sampled candidates and post-descent outcomes at selected m.
III. ALNS–quantum integration	Add quantum repair to ALNS's operator portfolio; compare hybrid against the same ALNS with classical-only repair under matched end-to-end budgets.	Final energy, escape rate, time, overhead and uncertainty estimates.
IV. Polyphase extension	Generalize classical operators and encoding to $Q = 4$, then consider larger Q and optional mismatched filtering.	Polyphase correctness tests, encoding/resource comparison and benchmark report.

Table 9: Proposed validation milestones; quantum and polyphase phases are future work.

11.1 System interface and reproducibility

The planned solver accepts a sequence length N, phase alphabet size Q, objective/constraint mode, algorithm selection, time budget, and random seed. It returns the phase sequence, unnormalized objective energy, merit factor when applicable, elapsed time, solver metadata and a validation report. The quantum module additionally records selected positions, Hamiltonian/QUBO description, logical-resource counts, measurement statistics and downstream post-relaxation outcomes. This hardware-agnostic interface can drive a classical simulator, quantum-circuit backend or quadratic-Ising optimization service.

Repository and reproducibility. Supporting implementation, experiment configurations, and verified result exports are to be linked from `https://github.com/Eulerq-Software/Vernex-Challenge-Submission`. The main dossier is intended to be readable without code access. Before submission, the repository URL, access policy, branch/commit, licensing and any confidential artefacts should be checked by the team.

12 Conclusion

We propose *ALNS with quantum-assisted neighbourhood repair* as a practical algorithmic direction for phase-coded radar waveform design. Instead of assuming that large LABS instances should be solved wholly on a quantum processor, the approach gives global optimization and reliability to classical ALNS while assigning selected, bounded-size subproblems to quantum sampling or optimization. This responds directly to the auxiliary-variable overhead of full QUBO reductions and the nonlocal circuit cost of the higher-order LABS Hamiltonian.

The proposal is grounded in completed classical research. Our baseline solvers and landscape experiments reveal difficult local minima, productive uphill triggers and descent cascades; sequential guidance improves basin-escape success in several controlled comparisons. These results motivate the design of quantum candidate generators whose merit is judged *after* classical relaxation and full-energy evaluation. Both higher-order QAOA and exact QUBO-based quadratic-Ising execution have clear mathematical roles in that design.

The intended outcome is an efficient and scalable hybrid solver, but this remains a *hypothesis to test*, not an established quantum speedup. The key next experiment compares the same ALNS framework with classical versus quantum neighbourhood repair using identical incumbents, budgets and postprocessing, including all quantum execution overhead. The same architecture also provides a systematic route to polyphase waveforms ($Q > 2$).

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